

Report
Of
"Data Science Master Virtual Internship"

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By
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Batch: 2026-27

CANDIDATE'S DECLARATION

I, Tushar Attri, do hereby declare that the internship report titled "Data Science Master Virtual Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Tushar Attri

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the AICTE - EduSkills Data Science Master Virtual Internship Program, supported by Siemens. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to the All India Council for Technical Education (AICTE), the National Internship Portal, EduSkills Foundation and Siemens Industry Software (India) Pvt. Ltd. for providing this valuable learning opportunity and for sharing their truthful and illuminating views on a number of issues related to this work.

I would also like to thank my faculty guide and the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement throughout the internship and the preparation of this report.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Tushar Attri

Roll No.: 25SCS1003000538

INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Tushar Attri

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 497fac1ca5c216943b93

Student ID: 25SCS1003000538



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-Week AICTE - EduSkills Virtual Internship Program undertaken in the domain of "Siemens Data Science Master", carried out during June 2026 to August 2026 (AICTE Student ID: 25SCS1003000538). The internship was pursued as part of the Bachelor of Technology (Computer Science and Engineering) curriculum at IILM University, Greater Noida, under the National Internship Portal initiative of the Ministry of Education, in association with the All India Council for Technical Education (AICTE), EduSkills Foundation and Siemens Industry Software (India) Pvt. Ltd.

The internship was designed to provide structured, industry-relevant learning in the field of Data Science and Machine Learning using the RapidMiner platform, combining self-paced online modules with weekly assessments, project documentation, and a final assessment test. This report documents the organization behind the program, the problem it addresses, the objectives and scope of the learning undertaken, the tools and technologies used, the underlying platform architecture, the methodology followed, and the outcomes achieved.

4.2 Organization Profile

The internship was jointly conducted and supported by the following organizations:

- All India Council for Technical Education (AICTE): The statutory body of the Government of India responsible for planning and coordinated development of technical education, functioning under the Ministry of Education. AICTE oversees the National Internship Portal through which this virtual internship was offered.
- EduSkills Foundation: A skilling initiative working under the theme "Nation Building Through Skills", EduSkills partners with global technology companies to design and deliver industry-aligned virtual internship programs to engineering and technology students across India.

- Siemens Industry Software (India) Pvt. Ltd.: The industry partner supporting the Data Science Master track, providing curriculum content, certification and mentorship through its Altair/RapidMiner data science tools.
- RapidMiner (an Altair Company): The data science and machine learning platform used throughout the internship for hands-on learning, covering data engineering, machine learning, application deployment and platform administration.
- IILM University, Greater Noida: The host institute, School of Computer Science and Engineering, under whose academic supervision this internship and report were undertaken as part of the B.Tech CSE curriculum.

4.3 Problem Statement

With the rapid adoption of Artificial Intelligence and Machine Learning across industries, there exists a significant gap between academic curricula and the practical, tool-based skills expected by employers in the data science domain. Students often possess theoretical knowledge of statistics and algorithms but lack hands-on exposure to industry-standard platforms used for data access, transformation, model building, deployment and enterprise-level platform administration. This internship was designed to address that gap by providing a structured, week-wise, hands-on learning path using the RapidMiner platform - covering the complete data science lifecycle from business problem identification to model deployment and monitoring - thereby making participating students industry-ready.

4.4 Project Objectives

- To identify business problems that are suitable for solving using Artificial Intelligence and Machine Learning techniques.
- To learn how to access, load, transform and calculate data using industry-standard data engineering practices.
- To build foundational and advanced machine learning models, including classification, regression, clustering and association rule mining.
- To master advanced data handling, automation and text processing techniques.
- To learn advanced machine learning techniques such as parameter optimization, cross-validation, feature engineering and time-series forecasting.

- To deploy and monitor machine learning models and understand model management and web application integration.
- To configure and manage the RapidMiner AI Hub, including enterprise-level platform operations, Radoop, real-time scoring and the RapidMiner Marketplace.
- To successfully clear weekly assessments, project documentation submissions and the Final Assessment Test to earn the Virtual Internship Certificate.

4.5 Scope of the Project

The internship followed a structured 8-week, week-wise learning plan, with each week covering a specific module of the RapidMiner certification track, followed by a Final Credential Validation / Internship Grade Point Assessment:

<i>Week</i>	<i>Module</i>	<i>Description</i>
Week 1	Applications & Use Cases Professional	Identifying business problems suitable for AI and Machine Learning.
Week 2	Data Engineering Professional	Accessing, loading, transforming and calculating data.
Week 3	Machine Learning Professional	Building foundational machine learning skills.
Week 4	Data Preparation Master	Mastering advanced data handling and automation.
Week 5	Machine Learning Master	Learning advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploying and monitoring ML models.
Week 7	Platform Administration Master	Configuring and managing RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.

Note: This being a virtual, skill-based internship, there was no monetary stipend associated with it; the focus and scope remained entirely on structured learning, assessment and certification. Table 4.5.1: Week-wise scope of the Data Science Master Virtual Internship.

4.6 Technologies and Tools Used

- RapidMiner Studio - for visual, code-free design of data science workflows (data access, transformation, modelling).
- RapidMiner AI Hub - for enterprise deployment, scheduling, model management and collaboration.
- RapidMiner Radoop - for big-data integration with Hadoop-based environments.
- RapidMiner Real-Time Scoring - for deploying models as real-time scoring web services.

- RapidMiner Marketplace - for extending platform functionality with additional operators and extensions.
- Machine Learning techniques - Classification, Regression, Clustering, Association Rules, Cross-Validation, Feature Engineering and Time-Series Forecasting.
- Data handling techniques - Loops and Branches, Advanced Text Processing, Exception Handling, Logging, Data Cleansing, Regular Expressions, Macros, Web APIs and Scripting.
- CRISP-DM methodology - the industry-standard Cross-Industry Standard Process for Data Mining, used as the conceptual framework throughout the program.

4.7 System Architecture

The RapidMiner platform used during the internship follows a layered client-server architecture consisting of the following components:

- Client Layer - RapidMiner Studio, the visual workflow designer used by data scientists to build, test and validate data science processes on the local machine.
- Server/Management Layer - RapidMiner AI Hub, which centrally manages processes, schedules jobs, controls user access, and stores shared repositories, connections and projects.
- Big-Data Integration Layer - RapidMiner Radoop, which connects the platform to Hadoop/Spark clusters for processing large-scale datasets directly within the cluster.
- Deployment/Scoring Layer - Real-Time Scoring services, which expose trained models as REST web services that can be consumed by external applications for live predictions.
- Extensibility Layer - RapidMiner Marketplace, which provides additional operators, connectors and extensions that can be plugged into the core platform as required.

This layered architecture allows an end-to-end data science workflow - from raw data ingestion, through model building and validation, to enterprise-wide deployment and monitoring - to be managed within a single, unified platform.

4.8 Methodology

The internship curriculum was structured around the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, which was mapped across the eight weeks of learning as follows:

- Business Understanding - Week 1 (Applications & Use Cases Professional): identifying business problems suitable for AI/ML solutions.
- Data Understanding & Preparation - Weeks 2 and 4 (Data Engineering Professional and Data Preparation Master): accessing, loading, cleansing and transforming data.
- Modelling - Weeks 3 and 5 (Machine Learning Professional and Machine Learning Master): building, validating and optimizing predictive models.
- Evaluation & Deployment - Week 6 (Applications & Use Cases Master): deploying models into applications and monitoring their performance.
- Operations & Administration - Weeks 7 and 8 (Platform Administration Master): configuring and administering the RapidMiner AI Hub at an enterprise level.

Each week combined self-paced video learning with hands-on exercises, followed by a weekly assessment. Selected modules also required submission of project documentation. In the final week, all learning culminated in a Final Assessment Test, and an overall grade was computed based on weekly assessments, project submissions and the final test score.

4.9 Expected Outcomes

On successful completion of the 8-week Data Science Master Virtual Internship, the following outcomes were achieved:

- Successfully cleared all weekly assessments, project submissions and the Final Assessment Test.
- Earned individual RapidMiner certifications at the Professional and Master levels across Applications & Use Cases, Data Engineering and Machine Learning, along with Platform Administration Master certification.

- Awarded the AICTE-EduSkills Certificate of Virtual Internship for the Data Science Master track, supported by Siemens, with an overall Internship Grade of "O" (Outstanding), as verified by Certificate ID 497fac1ca5c216943b93 / Student ID 25SCS1003000538.
- Gained practical, industry-relevant proficiency in the complete data science lifecycle - from business problem identification and data engineering to machine learning, model deployment and enterprise platform administration.
- Strengthened readiness for entry-level roles and further academic/industry pursuits in Data Science, Machine Learning and AI-driven application development.

4.10 Certificates of Completion

The following certificates were issued in recognition of successful completion of each module of the Data Science Master Virtual Internship. The overall Internship Completion Certificate has already been presented in Section 3 of this report.

Applications & Use Cases - Professional Certification



Data Engineering - Professional Certification




This is to certify that

Tushar Attri

successfully passed the examination for
Data Engineering Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:
Data Access, Basic Transformations, Working with Multiple Data Sets, Pivot Tables, Routines and Simple Text Processing.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.



Knut Makowski
(Head of Education)

Machine Learning - Professional Certification




This is to certify that

Tushar Attri

successfully passed the examination for
Machine Learning Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:
Classification, Regression, Scoring, Hold-out Validation, Correlations, Feature Importance, Clustering and Association Rules.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.



Knut Makowski
(Head of Education)

Data Engineering - Master Certification




This is to certify that

Tushar Attri

successfully passed the examination for
Data Engineering Master Certification

The Master level is the second level after Professional and stands for high proficiency in the topic. Some of the topics tested at this level are:
Loops and Branches, Advanced Text Processing, Exception Handling, Logging, Data Cleansing, Regular Expressions, Macros, Web APIs, and Scripting.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.



Knut Makowski
(Head of Education)

Machine Learning - Master Certification



Applications & Use Cases - Master Certification



Platform Administration - Master Certification



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MAIL PROOF

